

FAKHUR U DIN

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RESEARCH DIRECTION / INTEREST

My work centers on practical, applied Machine Learning — building systems that automate the data-to-decision pipeline (preprocessing, model training, prediction, and explainability), as well as recommendation systems. I am interested in extending this into graduate research on AutoML and explainable AI (XAI).

ACADEMIC QUALIFICATION

BS Information Technology

Expected: January 2027

Shaheed Benazir Bhutto University, Shaheed Benazirabad (SBA) | CGPA: ~3.2 / 4.0

RESEARCH & PROJECTS

Intelligent Data Analysis System Using AI Agents (Final Year Project)

The Intelligent Data Analysis System using AI Agents is a web-based platform that automates the complete data analysis process, from preprocessing to prediction. It allows users to upload datasets and automatically performs data cleaning, EDA, and machine learning. With AutoML and explainable AI, the system converts raw data into meaningful insights through a simple one-click pipeline.

- Automated Preprocessing: handles missing values, duplicates, outliers, and data types
- Smart EDA: generates summaries, trends, and correlation analysis
- AutoML: detects problem type, trains multiple models, and selects the best one
- Model Explainability: feature importance and confusion matrix for transparent results
- Interactive Dashboard: real-time results, data export, and automated reporting

Interactive Data Visualization Web App

Built an interactive Data Visualization App using Python, Streamlit, and Seaborn. Users can upload datasets, explore insights, and create dynamic plots (scatter, bar, histogram, box, pair, facet grid). The app features real-time customization, session-state logic, and a modern CSS-styled UI. Focused on clean, maintainable code, responsive design, and smooth user experience.

Tools: Python, Streamlit, Seaborn, pandas, CSS

AI-Powered Book Recommendation System

Developed an intelligent book recommendation system using Python. Implemented KNN with cosine similarity and Truncated SVD to improve accuracy and performance. Designed an interactive Streamlit web app that suggests similar books based on a user's selected title with appealing visual output.

Tools: Python, Scikit-Learn, pandas, NumPy, Streamlit

ACADEMIC PROJECTS

Tailor Management System, Real Estate Agency Management System, Text Editor, and File Sorter — desktop applications built with Python and Tkinter as part of coursework.

TECHNICAL SKILLS

Languages & Tools: Python, Tkinter **Data & ML Libraries:** pandas, NumPy, Scikit-Learn, Seaborn **Frameworks:** Streamlit

CERTIFICATIONS & ACHIEVEMENTS

- Hands-on Training in Python Development — PITP Phase-II, Government of Sindh, Sukkur IBA University, Shaheed Benazirabad (Sept 22 – Nov 16, 2025)
- 2nd Position, Tech Exhibition Competition — Pakistan Independence Day Celebrations, Shaheed Benazir Bhutto University, Shaheed Benazirabad (August 2025)

REFERENCES

Dr. Mairaj Nabi

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Dr. Asif Ali Wagan

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